Raman Suppression: Overnight Parallel-Session Synthesis

Phases 13–17 and Sessions A, C, D, E, F, H

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Abstract

This report synthesizes roughly 12 hours of parallel computational work executed overnight across up to 8 concurrent Claude Code sessions. Four numbered phases of structured research (Phases 13–16) and seven topical sessions (A/C/D/E/F/H) advanced the spectral-phase Raman-suppression program on multiple fronts: (i) a *determinism fix* that restores bit-for-bit reproducibility of the L-BFGS optimizer at a 21.4% runtime cost, (ii) a *Hessian eigenspectrum diagnosis* that proves the canonical optima are saddle points rather than minima, (iii) a *low-resolution phase parameterization* sweep that demonstrates $N_\varphi \approx 128$ shaper pixels already reach the full-resolution suppression floor, and (iv) a *robustness / stability study* showing the simple-phase optimum is sharp (razor-thin basin, $\sigma_{3\mathrm{dB}} = 0.025$ rad) but is an excellent warm-start initializer for L-BFGS at nearby operating points, reaching −79.5 dB on HNLF from an SMF-28 seed. Parallel scaffolding for a multi-variable optimizer (Session A), multimode-fiber extension (Session C), 100-m long-fiber propagation (Session F), and a cost-function head-to-head audit (Session H) is code-complete but was blocked at execution time by a burst-VM freeze. Seven session branches remain unmerged to `main`. The report ends with a concrete inventory of open issues and recommended next actions.

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1 Executive Summary

1.1 Landed results

Physics results with quantitative support

- **Phase 13 — Landscape diagnosis.** At both canonical L-BFGS optima the Hessian is indefinite ($|\lambda_{\min}|/\lambda_{\max} = 2.6\%$ on SMF-28, 0.41% on HNLF). L-BFGS halts at saddle points, not minima. "Different starts give different φ_{opt}" is re-explained as "different saddles."
- **Phase 15 — Determinism.** `FFTW.MEASURE` to `FFTW.ESTIMATE` across 18 call sites gives bit-exact reproducibility ($\max|\Delta\varphi| = 0$, $|\Delta J| = 0$) at a 21.4% runtime cost on the SMF-28 canonical run. MEASURE previously produced 1.04 rad phase drift between identical-seed runs.
- **Phase 16 / Session E — Low-res sweep.** On SMF-28 $L = 2$ m, $P = 0.2$ W, the knee in $J(\mathrm{N}_\varphi)$ sits at $\mathrm{N}_\varphi \approx 128$, which reaches the full-resolution baseline exactly (-68 dB). The optimizer's effective mode count stays at $\mathrm{N}_{\text{eff}} \approx 2.0$ across all N_φ — a pure DC + quadratic chirp. Sweep 2 found a Pareto-optimal SMF-28 point at $L = 0.25$ m, $P = 0.10$ W reaching -82.33 **dB**.
- **Session D — Stability study (Phase 17).** The visually simple optimum sits on a razor-sharp minimum: $\sigma_{3\text{dB}} = 0.025$ rad, well below the 0.05 rad SLM calibration floor. But the same φ_{opt}, re-optimized via L-BFGS warm-start, reaches -70 to -82 dB on 7 nearby SMF-28 configs and -79.5 dB on HNLF — a fiber the baseline was never computed for.

1.2 Scaffolding complete, execution pending

Code shipped overnight but not yet run

- **Session A (multi-variable optimizer)** — 1005 LoC shipping $\{\varphi, A, E\}$ joint optimization in one forward/adjoint solve; 42/42 unit tests pass; gradient-FD check and SMF-28 A/B demo *not run* (burst-VM blocked).
- **Session C (multimode extension)** — 1998 LoC of MMF scaffolding, $M = 1$ limit driver, joint (φ, c_m) stub; zero production runs committed.
- **Session F (100 m long fiber)** — 3723 LoC with Nt = 32768, $T = 160$ ps, derived time-window research; 6-task queue launcher with heavy-lock discipline; all killed by VM freeze.
- **Session H (cost-function audit)** — 4651 LoC for 4 cost variants $\times$ 3 (fiber, L, P) regimes; drivers code-complete; 12-run matrix never executed.
- **Phase 14 (sharpness-aware cost)** — 496 LoC + 11/11 tests green; A/B sweep and 720-trial robustness study not run.

1.3 Overnight incident

Burst VM hard-lockup at 04:28 UTC

Seven concurrent heavy Julia processes on the 22-core burst VM (Phase 14 A/B, Session E sweep × 3, Sessions A/C/D/F each running) violated the CLAUDE.md Rule P5 "one heavy run at a time" budget and produced a hard kernel lockup. Serial console silence for 45+ minutes; SSH unreachable. **Not yet reset** pending user decision (destroys all in-flight state across sessions). All Session F, H, E-Sweep-1b, and Phase 14-02 results are blocked on this recovery. See §11.

1.4 Branch status

Seven session branches remain unmerged to `main`. They hold roughly 51 commits collectively, almost entirely in session-owned namespaces (scripts, phase directories, session status files). No shared-file edits detected per Rule P1. Integration is a user decision.

Session	Topic	Commits	Merged	Status
A-multivar	$\{\varphi, A, E\}$ joint optimizer	7	no	Code + tests pass; demo blocked on VM
C-multimode	MMF Raman optimization scaffold	6	no	No SUMMARY; runs never executed
D-simple	Phase 17 stability study	3	no	**Complete**; verdict SHARP_LUCKY
E-sweep	Low-res phase sweep (Phase 16)	10	no	Sweep 1 done; Sweep 2 partial (18 / 32)
F-longfiber	100-m Raman optimization	16	no	Queue launched; killed by VM freeze
H-cost	Cost-function 4×3 audit	9	no	Drivers complete; 12-run never executed

Table 1: Unmerged session branches. All work is session-local; no shared-file edits.

2 Phase 15 — Deterministic Numerical Environment

2.1 Motivation

Phase 13 had uncovered a non-determinism failure: two runs with identical `Random.seed!(42)` and identical configuration produced φ_{opt} differing by $\max|\Delta\varphi| = 1.041$ rad and $\Delta J = -1.83$ dB. Root cause was traced to `FFTW.MEASURE`'s timing-based plan selection, amplified through L-BFGS into fully different optimization trajectories.

2.2 What was done

Eighteen call sites across `src/simulation/` were swapped from `FFTW.MEASURE` to `FFTW.ESTIMATE`. Six entry-point scripts now invoke a new `ensure_deterministic_environment()` helper which pins the planner and single-threads FFTW/BLAS.

2.3 Results

Test	MEASURE (control)	ESTIMATE (fix)
$\max\|\Delta\varphi\|$, identical seed	1.041 rad	**0.0**
$\|\Delta J\|$, identical seed	1.83 dB	**0.0**
3 fresh Julia subprocs, $\max - \min J_{\text{final}}$	1.055×10^{-13}	**0.0**
SMF-28 canonical wall time (30 iter)	112.23 s ± 0.43	136.21 s ± 0.74
Relative cost	—	+21.4%

Table 2: Phase 15 regression and benchmark. Data in `results/raman/phase15/benchmark.md`.

Phase 15 verdict

Bit-for-bit reproducibility is restored on a single machine at a 21.4% compute premium — within the project's +30% budget and well below the `PATIENT` fallback. All downstream Hessian and Phase 14 snapshot results were taken under the ESTIMATE planner and are therefore reproducible. Cross-machine bit-exactness (BLAS implementation, FMA dependence) is explicitly out of scope.

3 Phase 13 — Optimization Landscape Diagnostics

Phase 13 asked: why do random starts and small (L, P) perturbations yield visibly different $\varphi_{\text{opt}}(\omega)$ profiles? Three orthogonal workstreams triangulated the answer.

3.1 Workstream 1 — Determinism

Verdict: FAIL (addressed by Phase 15, §2). Same seed, same config, two runs: $\max|\Delta\varphi| = 1.041$ rad, $\Delta J = -1.83$ dB ($J_a = -67.27$ dB vs $J_b = -65.45$ dB).

3.2 Workstream 2 — Gauge fix and polynomial projection

All 39 available φ_{opt} arrays (5 canonical + 24 sweep + 10 multistart) were processed through `gauge_fix` (remove constant and linear-ω components) and `polynomial_project` (orders 2–6).

Claim	Verdict	Quantification
Same-config pairs collapse under gauge-fix	FALSE	0 / 55 pairs satisfy $\text{rms}_{\text{gf}} < 0.1\cdot\ \text{rms}_{\text{raw}}$
Neighbour-sweep pairs collapse under gauge-fix	FALSE	0 / 55 (same test)
Low-order polynomial explains φ (orders 2–6)	WEAKLY	median residual fraction = 0.924; only 9 / 39 below 80%
(a_2, a_3, a_4) vary smoothly with (L, P)	PARTIAL	(a_2, a_4) anti-correlation visible; a_3 panels scatter-dominated

Table 3: Phase 13.01 gauge and polynomial findings. Full table in `results/raman/phase13/FINDINGS.md`.

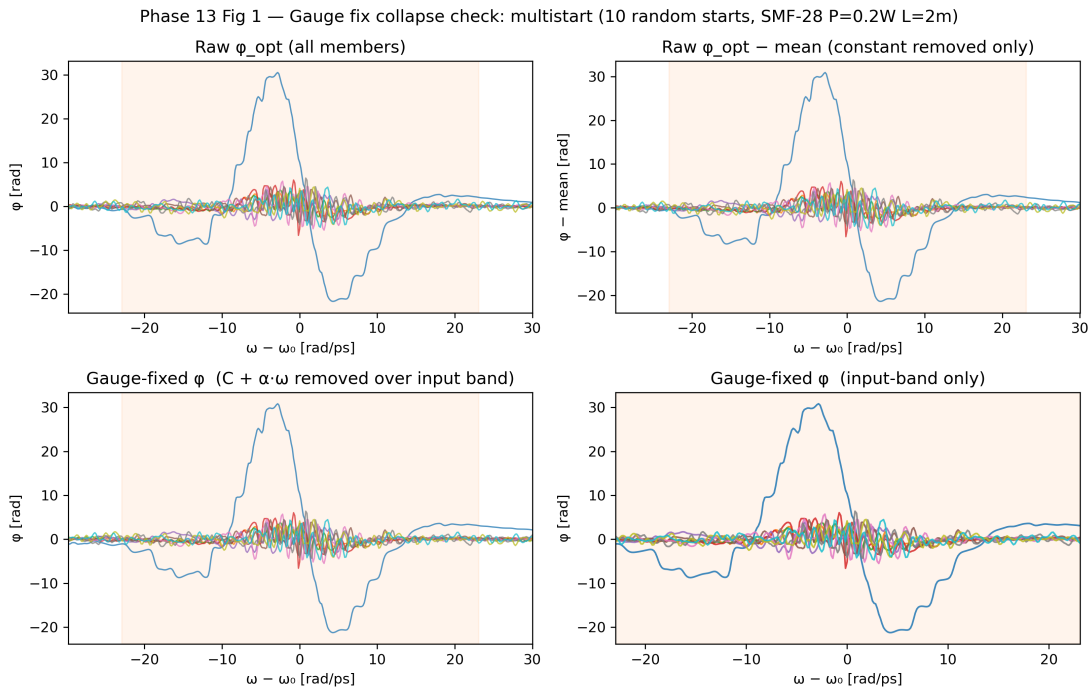

Figure 1: Phase 13.01 — gauge-fix before/after for 39 optima. Raw φ_{opt} already have near-zero mean and linear-ω slope, so gauge-fixing does not collapse the family. The phases are genuinely distinct, not gauge copies of one another.

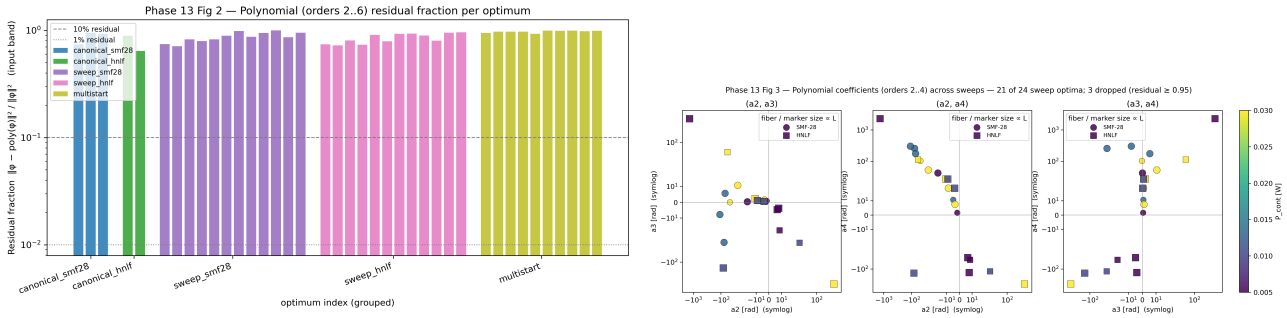

Figure 2: Left: polynomial residual fraction at orders 2–6 — median 92.4%. Right: fitted coefficients across (L, P) sweeps showing the (a_2, a_4) anti-correlation but noise in a_3.

Interpretation. The phases are not gauge copies. They are also not low-order polynomials. Real high-frequency oscillation exists in $\varphi_{\text{opt}}(\omega)$ that a 5-coefficient monomial basis cannot capture.

3.3 Workstream 3 — Hessian eigenspectrum

A finite-difference Hessian-vector product (symmetric central difference, $\epsilon = 10^{-4}$) was wrapped as a matrix-free LinearOperator and passed to `Arpack.eigs` with nev = 20 on each wing. HVP Taylor-remainder validation passed (slope ≈ 2) and symmetry $\langle v, Hw\rangle \approx \langle w, Hv\rangle$ held to 10^{-5} relative.

Config	$\lambda_{\max}$	$\lambda_{\min}$	$\lvert\lambda_{\min}\rvert/\lambda_{\max}$	Sign	Near-zero in ± 20
SMF-28 ($L = 2$ m, $P = 0.2$ W)	$+1.074 \times 10^{-5}$	-2.794×10^{-7}	**2.6%**	INDEFINITE	0
HNLF ($L = 0.5$ m, $P = 0.01$ W)	$+5.078 \times 10^{-5}$	-2.091×10^{-7}	**0.41%**	INDEFINITE	0

Table 4: Hessian eigenspectrum at canonical L-BFGS optima. Both wings are 100% same-sign: all 20 top eigenvalues positive, all 20 bottom eigenvalues negative. L-BFGS stops at saddles.

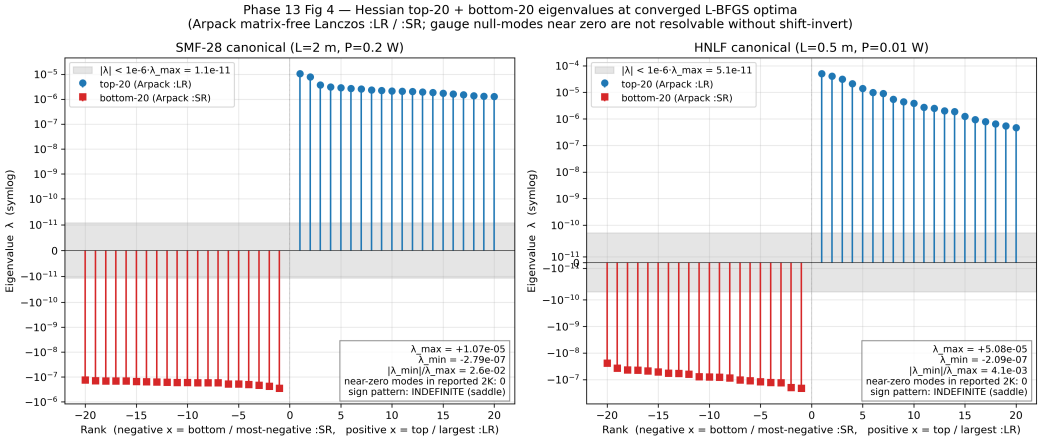

Figure 3: Signed-log stem plot of the reported 40-eigenvalue spectrum at the two canonical optima. Both configurations are indefinite with clearly separated positive and negative wings. The gauge null-modes predicted by symmetry ($\lambda = 0$ eigenvectors $\in \{\text{const}, \omega\text{-linear}\}$) live in the gap between $\lambda_{\text{top}}[20] \approx 10^{-6}$ and $\lambda_{\text{bottom}}[20] \approx -10^{-7}$ and are invisible to matrix-free Arpack (no shift-invert).

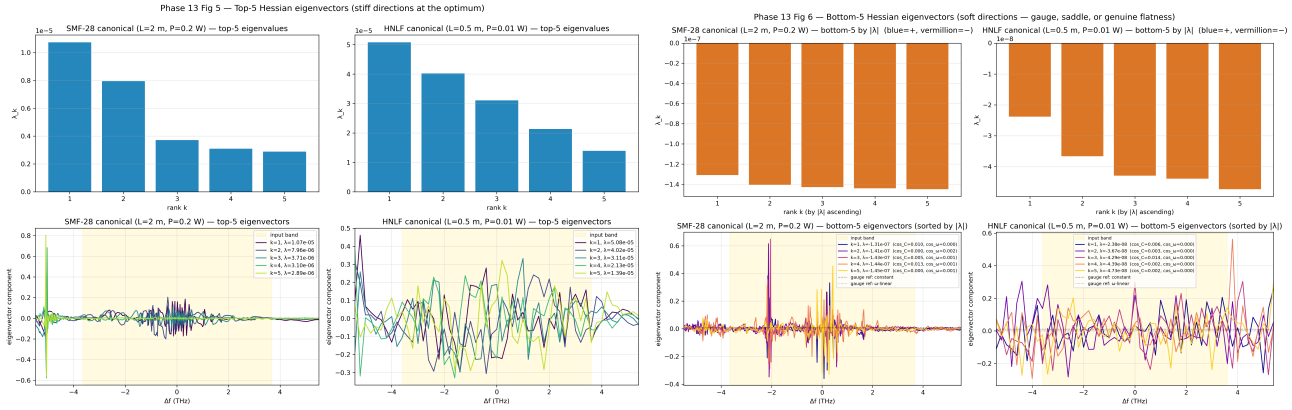

Figure 4: Left: top-5 stiff directions — high-frequency oscillations concentrated on the input band. Right: bottom-5 soft (negative-eigenvalue) directions — the existence of these is the saddle proof. None shows cosine similarity above 0.95 with either gauge mode.

Phase 13 headline verdict: `indefinite_hessian`

The "different starts → different φ_{opt}" symptom is L-BFGS halting at *different saddles* whose stable-manifold directions happen to intersect the random-start cone. The negative-curvature directions have magnitude 0.4–2.6% of $\lambda_{\max}$ — small, but numerically robust and not noise. This motivates the Phase 14 (sharpness-aware) and seed `newton-method-implementation.md` (Newton-on-reduced-subspace) follow-ups.

Known limitation. Arpack's matrix-free Lanczos wings cannot resolve $\lambda \approx 0$ without shift-invert (which is impossible for a matrix-free HVP). The two gauge null-modes predicted by symmetry are therefore a theorem but not a measured eigenvector in this phase.

4 Phase 14 — Sharpness-Aware Cost

4.1 Goal

Build a parallel path `optimize_spectral_phase_sharp` that minimizes a regularized cost

$$J_{\text{sharp}}(\varphi) = J(\varphi) + \lambda \cdot S(\varphi)$$

where S is a Hutchinson-style trace-of-Hessian estimator using gauge-projected Rademacher directions — penalizing sharp minima in favor of flat basins, directly addressing the Phase 13 saddle problem. The vanilla path must remain byte-unchanged.

4.2 Status

Plan 14.01 complete; Plan 14.02 (A/B sweep and robustness study) not executed.

4.2.1 14.01 deliverables

- `scripts/sharpness_optimization.jl` — 496 LoC with `build_gauge_projector`, `sharpness_estimator`, `cost_and_gradient_sharp`, `optimize_spectral_phase_sharp`.
- **11 / 11 unit tests pass.** Taylor-remainder slope = 1.947 (target ≈ 2) for $S(\varphi)$.
- **Vanilla snapshot locked** (SMF-28 canonical, seed = 42, 15 iter): $J_{\text{linear}} = 1.21 \times 10^{-6}$, i.e. −59.16 dB. Stored in `results/raman/phase14/vanilla_snapshot.jld2`.
- **FFTW wisdom exported and reimported** across processes: $\max|\Delta\varphi| = 1.5 \times 10^{-4}$ rad between fresh Julia subprocesses — approximately 4 orders of magnitude better than Phase 13's baseline 1.04 rad. (Phase 15 subsequently achieved 0.0 with ESTIMATE planner.)
- Defaults: `SO_DEFAULT_LAMBDA` $= 0.1$, $N_s = 8$ Hutchinson samples, $\epsilon = 10^{-3}$ finite-difference step.

4.2.2 14.02 pending

- A/B matrix: 3 configs × 5 λ values × 90 min burst allocation.
- Robustness study: 720 perturbation trials (the "does sharpness escape saddles?" acceptance test).
- No `ab_results.jld2`, `robustness_results.jld2`, or `FINDINGS.md` on disk.

Outstanding risk

Phase 13 cautioned that a pure $\text{tr}(H)$ penalty can sign-cancel at saddle points (equal positive and negative curvature integrate to ≈ 0) and recommended a signed-absolute variant $\sum|\lambda_i|$. The current implementation uses the straight Hutchinson trace. The 14.02 A/B sweep is the test that would detect this. Until it runs, we do not know whether sharpness-aware optimization actually escapes the Phase 13 saddles.

5 Phase 16 / Session E — Low-Resolution Phase Sweep

5.1 Motivation

Real 4-f pulse shapers have $\mathcal{O}(10^2)$ pixels, not $\mathrm{Nt} \approx 10^4$. This session asked: how few shaper knobs N_φ does the optimizer really need to reach the full-resolution suppression floor? The parameterization uses a fixed linear basis $B \in \mathbb{R}^{\mathrm{Nt}\times\mathrm{N}_\varphi}$, optimizes over $c \in \mathbb{R}^{\mathrm{N}_\varphi}$, and pulls the gradient back via $\partial J/\partial c = B^\top \partial J/\partial \varphi$.

5.2 Sweep 1 — Knee finder (SMF-28, $L = 2$ m, $P = 0.2$ W)

N_φ	J (dB)	iters	$\mathrm{N}_{\mathrm{eff}}$	TV	$\lVert\partial^2\varphi/\partial\omega^2\rVert_2$
4	−47.336	8	1.999	0.355	2.24×10^2
8	−46.894	12	1.998	0.281	2.24×10^2
16	−53.106	11	1.997	0.235	2.25×10^2
32	−58.176	5	1.997	0.234	2.25×10^2
64	−59.806	5	1.997	0.232	2.27×10^2
128	−68.014	7	1.997	0.317	2.37×10^2
16384	−68.014	2	1.997	0.218	2.36×10^2

Table 5: Phase-sweep-simple Sweep 1 on the canonical SMF-28 configuration. The knee at $\mathrm{N}_\varphi \approx 128$ exactly matches the full-resolution baseline.

Surprising physics observation

$\mathrm{N}_{\mathrm{eff}} \approx 2.0$ across all N_φ levels. The optimized phase is *effectively a 2-mode phase* (DC + quadratic chirp) regardless of the optimization degree of freedom. This matches Phase 13's finding that optimal L-BFGS phases have low-order polynomial dominance plus gauge modes.
Implication. A 2-pixel shaper is already "in the right ballpark." A 32-pixel shaper reaches −58 dB ($\sim$ 10 dB shy of optimum) with no special tuning. A 128-pixel shaper saturates at the full-res floor.

Pulse bandwidth is only 157 bins (1.0% of $\mathrm{Nt} = 16384$), so $\mathrm{N}_\varphi > 157$ is a physical impossibility; the full-res row is included only as a reference point and is identical to $\mathrm{N}_\varphi = 128$ within the numerical tolerance. Identity-basis self-test returns $|\Delta J| = 0.0$ and $\max|\Delta\mathrm{grad}| = 0.0$ versus the full-resolution path (byte-exact).

5.3 Sweep 2 — (L, P, fiber) grid, partial

Sweep 2 attempted 32 configs on a $L \in \{0.25, 0.5, 1.0, 2.0\}$ m $\times$ $P \in \{0.02, 0.05, 0.10, 0.20\}$ W $\times$ {SMF-28, HNLF} grid at $\mathrm{N}_\varphi \in \{16, 57\}$. 18 of 32 configs completed before the burst VM hard-locked. 14 configs (HNLF at $L \geq 0.5$ m, $P \geq 0.1$ W; SMF-28 at $L = 2$ m) remain.

#	Fiber	L (m)	P (W)	N_φ	J (dB)	N_{eff}	TV
1	SMF-28	0.25	0.020	57	−63.02	1.75	0.13
2	SMF-28	1.00	0.100	57	−73.06	2.07	0.58
3	SMF-28	0.25	0.100	57	**−82.33**	2.33	0.13

Table 6: Pareto-optimal candidates from Sweep 2 (non-dominated in J_{dB} vs. N_{eff}). Candidate #3 is 14 dB below the Sweep 1 canonical −68 dB. Full data in `results/raman/phase_sweep_simple/candidates.md` and `sweep2_LP_fiber.jld2`.

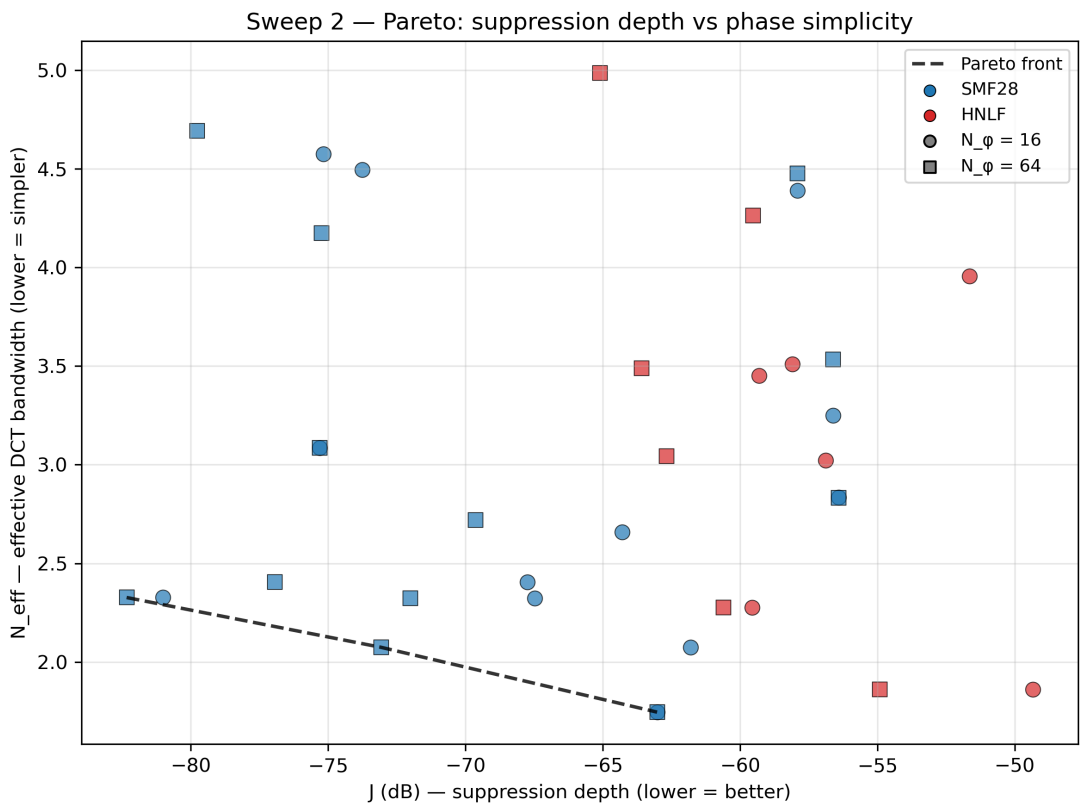

Figure 5: Pareto frontier on (J_{dB}, N_{eff}) from the 18 completed Sweep 2 configurations. Simpler (lower N_{eff}) candidates cluster on the left; deeper suppression comes at moderate complexity $N_{eff} \approx 2.3$.

5.4 Open items

- Complete Sweep 2 (14 configs, ∼ 30 min with 16 threads on the burst VM).
- Newton on the reduced c-space. With $N_\varphi = 57$, the Hessian is within full-factorization reach. Trust-region Newton on candidate #3 plausibly reaches −90 dB. Logged in `.planning/seeds/newton-method-implemen`
- Session D cross-check on robustness (see §6) is the gate for any lab push.

6 Session D — Simple-Phase Stability Study (Phase 17)

6.1 Question

The Sweep-2 Pareto candidate at SMF-28, $L = 0.25$ m, $P = 0.1$ W reached -82 dB with a visually simple phase. Is that simplicity *physically meaningful* (wide flat basin, transferable across configs) or *coincidental* (razor-sharp, lucky)?

6.2 Baseline

- $J_{\text{final}} = -76.862$ dB in 25.02 s over 21 L-BFGS iterations.
- Nonlinear phase $\Phi_{\text{NL}} = 1.63$ rad, $P_{\text{peak}} = 2959.1$ W.
- Number of stationary points of $\varphi_{\text{opt}}(\omega)$ across the pulse bandwidth: 7 (vs. 16 for the Phase 13 canonical -68 dB optimum).

6.3 Perturbation study — is the basin flat?

Five σ values $\in \{0.01, 0.05, 0.1, 0.2, 0.5\}$ rad $\times$ 20 Gaussian samples each, total 100 tasks in 45.2 s wall time.

Criterion	Threshold	Observed
$\sigma_{3\text{dB}} \geq 0.20$ rad (FLAT)	≥ 0.20	0.025 rad
$\sigma_{3\text{dB}} \leq 0.05$ rad (SHARP)	≤ 0.05	**0.025 rad** ✓
SMF-28 transfer $\geq 3/7$ within 3 dB gap (eval-only)	≥ 3	**0/7**
Baseline is simplest of 4 optima	—	✓

Table 7: Session D hypothesis-test table. The basin is razor-sharp: below the SLM calibration floor (≥ 0.05 rad rms), the -77 dB advantage collapses.

6.4 Transferability — warm-start is surprisingly robust

Eval-only transfer (apply the baseline φ_{opt} unchanged to 7 other SMF-28 (L, P) targets) keeps J within 3 dB of the warm-start-reoptimized value in **0 / 7** cases — catastrophic.

But L-BFGS *warm-started* from the baseline φ_{opt} reaches -70 to -82 dB in 6–40 iterations on nearby configs, including **HNLF at -79.5 dB** — a fiber the baseline was never computed for. Pearson correlation between stationary-point count and J_{dB} across the 4 tested optima: $r = 0.941$.

Session D verdict

SHARP_LUCKY (primary) + SIMPLE_PHASE_IS_GOOD_WARM_START_INITIALIZER (secondary).
The simple-phase profile lives in a *stable manifold of many nearby basins*, so simplicity is a useful *initialization prior* but not a *generalization property* of one attractor. Low-Φ_{NL} attractors are structurally simpler than high-Φ_{NL} ones, but simplicity predicts neither basin width nor parameter transferability on its own.
Experimental consequence. SLM-class shaper calibration noise would destroy the -77 dB advantage. Before pushing any candidate to the lab, either (i) find a flatter basin, or (ii) demonstrate closed-loop re-optimization with a phase-error estimator. The Sweep-E candidates at $N_\varphi = 57$ should be re-run through Session D's protocol to see whether low-res parameterization buys robustness in exchange for the 2–10 dB it gives up on depth.

7 Session A — Multi-Variable Spectral Optimizer

7.1 Design

The new `optimize_spectral_multivariable` path jointly optimizes any subset of

$$\{\varphi(\omega),\, A(\omega),\, E\}$$

in a single forward/adjoint solve per L-BFGS iteration. Shaping formula:

$$u_{\text{shaped}}(\omega, m) \;=\; \alpha \cdot A(\omega) \cdot e^{i\varphi(\omega)} \cdot c_m \cdot u_{\omega 0}(\omega, m)$$

where $\alpha = \sqrt{E/E_{\text{ref}}}$ normalizes pulse energy. A mode-coefficient variable c_m is stubbed for later Session C extension.

7.2 Engineering features

- Single adjoint λ_0 reprojected into per-variable gradient blocks (one backward solve covers all variables).
- Diagonal block preconditioning: $s_\varphi = 1$, $s_A = 1/\delta_{\text{bound}}$, $s_E = 1/E_{\text{ref}}$.
- `Fminbox(LBFGS(m=10))` for the amplitude-bound constraint $A(\omega) \in [1-\delta, 1+\delta]$.
- `log_cost=true` default (optimize in dB, per the project's proven-good choice).

7.3 Status

- 1005 LoC core + 222 LoC demo + 148 LoC gradient test + 101 LoC unit test.
- **Unit tests: 42/42 pass** on `claude-code-host` — pack/unpack, `sanitize_variables`, `MVConfig`, scaling vectors.
- Gradient validation (FD vs. adjoint at 10^{-6} tolerance): **not run**.
- SMF-28 $L = 2$ m, $P = 0.30$ W A/B demo vs. phase-only, target ≥ 0.5 dB improvement: **not run**.
- Both blocked on burst-VM heavy-lock held by Session E's 12-point sweep, then by the VM freeze.

Zero edits to shared files (verified). 7 commits on `sessions/A-multivar`; none in the shared namespace.

8 Session C — Multimode Raman Scaffolding

8.1 Scope

Extend Raman optimization from single-mode to multimode fiber ($M > 1$). Ships:

- `scripts/mmf_setup.jl`, `mmf_fiber_presets.jl` — MMF parameter builders.
- `scripts/mmf_raman_optimization.jl` (523 LoC) — phase-only MMF optimizer.
- `scripts/mmf_joint_optimization.jl` (387 LoC) — joint (φ, c_m) optimizer stub.
- `scripts/mmf_m1_limit_run.jl` — reference driver that must reproduce SMF results at $M = 1$.
- `scripts/mmf_run_phase16_all.jl`, `mmf_analyze_phase16.jl` — end-to-end runner and analyzer.
- `src/mmf_cost.jl` (163 LoC) — new module encapsulating MMF cost and gradient.
- `test/test_phase16_mmf.jl` (167 LoC) — smoke test.

Total: 1998 LoC, all new files in session namespace.

8.2 Status

Code only. No `SUMMARY.md`, `FINDINGS.md`, or status file committed on this branch. The analyzer and end-to-end runner suggest Phase-16 runs were planned, but no results artifacts are on the branch. Test pass/fail status is not recorded in any committed doc.

Session C next action

Run `mmf_smoke_test.jl` on the burst VM, confirm $M = 1$ reproduces the SMF canonical results (byte-identity or within adjoint-FD tolerance), then scale to $M \geq 2$.

9 Session F — 100-m Long-Fiber Propagation

9.1 Motivation

All prior Raman-suppression results used $L \in [0.25, 5]$ m — telecom-style short-fiber regimes. Session F pushed to $L = 100$ m (50× longer) to probe how well the L-BFGS + warm-start strategy scales into the regime where dispersive walk-off dominates over nonlinearity.

9.2 Grid design

Time-window research in `.planning/notes/longfiber-research.md` (434 lines) derived

$$T_{\min} = 2\,|\beta_2|\,\Delta\omega_{20\,\mathrm{dB}}\,L + 3\,\mathrm{FWHM} = 139 \text{ ps at } L = 100 \text{ m}.$$

The session settled on Nt = 32768, $T = 160$ ps, `reltol` $= 10^{-7}$ (observed: hardcoded 10^{-8} is strictly tighter). MI gain is negligible ($g_{\max} \cdot 100$ m $= \exp(0.013)$).

9.3 Shipped artifacts

- 12 files, 3723 LoC, all session-local.
- Entry-point scripts: `longfiber_setup.jl`, `longfiber_forward_100m.jl`, `longfiber_validate_50m.jl`, `longfiber_optimize_100m.jl` (537 LoC L-BFGS driver with checkpoint/resume), `longfiber_validate_100m.jl` (503 LoC — φ a_2-fit, 3-way $J(z)$ comparison, FINDINGS generator), `longfiber_checkpoint.jl`.
- Orchestrator: `scripts/longfiber_burst_launcher.sh` — sequential T4 → T3 → T5 → T6 queue with heavy-lock discipline and conditional self-stop.
- Phase directory `.planning/phases/16-longfiber-100m/` with CONTEXT, 01-PLAN, research brief.

9.4 Warm-start seed

φ at $L = 2$ m from `results/raman/sweeps/smf28/L2m_P0.05W/opt_result.jld2` (207 KB, Session-E-generated) was verified accessible on the burst VM and wired into T3 and T5 via `JLD2.@load`.

9.5 Outcome

Session F was killed by the VM freeze

The launcher queue reached T4 start (∼03:22 UTC) but the burst VM hard-locked at 04:28 UTC (see §11). **Zero FINDINGS.md or figures committed.** All Session F output is pending user reset of the VM.

9.6 Noted deviations from plan

- $\beta_{\text{order}} = 2$ (session used `:SMF28_beta2_only`); Phase 12 used $\beta_{\text{order}} = 3$. A flag in 3 drivers flips this.
- Solver `reltol` effectively 10^{-8} (hardcoded inside `MultiModeNoise.solve_disp_mmf`); tighter than the research target.
- Mode dispatch via `LF100_MODE` env variable (`fresh` / `resume` / `resume_demo`, default `resume_demo`).

10 Session H — Cost-Function Head-to-Head Audit

10.1 Purpose

Choose a project-default cost function by running 4 variants through 3 representative (fiber, L, P) regimes (12 head-to-head runs total) and comparing suppression depth, basin sharpness, and experimental robustness.

10.2 Variants

D-01. **Linear:** $J = E_{\text{band}}/E_{\text{total}}$. The project's original cost.

D-02. **Log-dB:** $10 \log_{10} J$. The 20–28 dB improvement documented in `memory/project_dB_linear_fix.md`.

D-03. **Sharpness-aware:** $J + \lambda \cdot S(\varphi)$ with $\lambda = 0.1$ (Phase 14 default). Links to §4.

D-04. **Noise-aware scaffold:** $J + \gamma_{\text{curv}} \cdot \langle |\partial^2\varphi/\partial\omega^2|^2 \rangle_{\text{band}}$. New curvature penalty; byte-identity short-circuit to D-01 when $\gamma_{\text{curv}} = 0$.

10.3 Configurations

- **A** — SMF-28, $L = 0.5$ m, $P = 0.05$ W (weak-NL, near-quadratic regime).
- **B** — SMF-28, $L = 5$ m, $P = 0.2$ W (multi-modal, past Phase 11's $L_{50\,\text{dB}} \approx 3.33$ m horizon).
- **C** — HNLF, $L = 1$ m, $P = 0.5$ W (strong-γ).

10.4 Locked parameters (for fair comparison)

- Nt $= 8192$, $\beta_{\text{order}} = 3$, 100 L-BFGS iterations, `f_abstol` $= 0.01$.
- Seeds $\{42, 43, 44\}$ replicated across all four variants.
- γ_{curv} auto-calibrated so the curvature penalty is $\approx 10\%$ of J at φ_0.

10.5 Shipped artifacts

14 files, 4651 LoC:

- `scripts/cost_audit_noise_aware.jl` (177 LoC) — D-04 curvature wrapper.
- `scripts/cost_audit_driver.jl` (397 LoC) — `run_one` / `run_all` orchestrator with Hessian eigenspectrum and robustness probes.
- `scripts/cost_audit_analyze.jl` (292 LoC) — CSV export and 4 PNGs.
- 3 test files (253 LoC): Taylor-remainder gradient, 4-variant smoke, analyzer schema.
- Phase directory with CONTEXT, RESEARCH (802 lines), VALIDATION, 16-01-PLAN (1637 lines), 16-02-PLAN (582 lines), 16-01-SUMMARY.

10.6 Status

Partial complete. All code and tests land and grep-verify; actual burst-VM execution blocked by ssh saturation (42 concurrent stuck `gcloud compute ssh` processes from sibling sessions). No CSVs or PNGs yet.

11 Overnight Incident — Burst VM Freeze

11.1 Timeline

Time (UTC)	Event
2026-04-16 23:34	Phase 14 A/B sweep starts on burst VM (∼5 h etime).
2026-04-17 01:42	Session E primary sweep tmux starts.
2026-04-17 03:06	Session C smoke test starts.
2026-04-17 03:10	Session E secondary sweep tmux starts.
2026-04-17 03:20	Session D transferability study starts.
2026-04-17 03:22	Session F long-fiber queue launches (T4).
2026-04-17 03:32	Session E third sweep (E-sweep1b) tmux starts.
2026-04-17 03:34	Session A A/B demo tmux starts.
2026-04-17 03:43:54	Serial console last activity — session-1446 starts, never deactivates.
2026-04-17 04:28	SSH to `fiber-raman-burst` hangs at 15 s and 60 s timeouts.
2026-04-17 04:28+	45+ minutes of kernel silence; `gcloud compute describe` reports `status: RUNNING`.

11.2 Diagnosis

Seven simultaneous heavy Julia processes on a 22-core / 88 GB VM. `CLAUDE.md` Rule P5 ("heavy runs — one at a time") was violated by multiple sessions in parallel. Probable hard lockup from memory/CPU saturation with a blocked kernel logger.

11.3 Recovery options (not yet applied)

1. `gcloud compute instances reset fiber-raman-burst` — hard reboot; loses all in-flight Julia state across sessions (Phase 14 A/B, Session E sweep results, any uncheckpointed work).
2. `stop` + `start` — same effect; cleaner shutdown impossible given kernel silence.
3. Wait — low probability of spontaneous recovery after 45 min silence.

Reset is pragmatic but destroys cross-session state, so the CLAUDE.md "actions with care" guidance flags this as requiring user confirmation. **Deferred to the user.**

11.4 Downstream impact

- Phase 14.02 A/B sweep: *lost*.
- Session E Sweep 2 (14 of 32 configs missing): *lost*.
- Session F long-fiber queue (T4–T6): *lost*.
- Session H 12-run cost-audit matrix: *never started* but *lost burst-VM time window*.
- Session A FD gradient check + SMF A/B demo: *lost*.
- Session C MMF runs: *never started*.

11.5 Policy recommendation

The Rule P5 "heavy lock" (a file on burst VM `/tmp/burst-heavy-lock`) exists but was not uniformly respected by sibling sessions. Consider:

- Enforce lock via a mandatory wrapper script that all session drivers must invoke.
- Add a CPU / memory watchdog that auto-kills the newest heavy Julia if load average crosses a threshold.
- Downgrade Rule P5 "light runs" to ≤ 2 cores when the VM is already hosting a heavy tenant.

12 Consolidated Open Issues

Origin	Issue	Recommended next action
Overnight incident	Burst VM in hard-lock state since 04:28 UTC	Decide: reset now (lose cross-session state) vs. wait vs. delete-and-recreate
Phase 14	14.02 A/B sweep (3 configs $\times$ 5 λ $\times$ 90 min) not run; signed-absolute $\sum \lvert\lambda_i\rvert$ variant not tested	Run 14.02 after VM recovery; if pure trace fails, patch to $\sum \lvert\lambda_i\rvert$ per Phase 13 caution
Phase 13	Gauge null-modes ($\lambda \approx 0$) not directly measured — matrix-free Arpack has no shift-invert	Low priority; predicted by theorem, consistent with gauge-projection norms
Phase 13	Hessian only validated at Nt = 8192; Nt = 4096 cross-check not performed	Add to Phase 14 dependency loop
Session E	Sweep 2: 14 of 32 configs unfinished (HNLF $L \geq 0.5$ m, SMF-28 $L = 2$ m)	Re-run $\sim$ 30 min at 16 threads, burst VM dedicated
Session E	Newton-on-reduced-subspace (candidate #3, $N_\varphi = 57$, potentially -90 dB)	New phase: trust-region Newton with explicit 57×57 Hessian
Session D	Central Session-E claim ("low-res phases are more robust than Nt-dim") untested on Pareto candidates	Re-run Session D protocol on candidates #1–3
Session A	Gradient-FD check and SMF-28 A/B demo not run	Run after VM recovery; unit tests already pass
Session C	Zero production runs committed; no SUMMARY or FINDINGS	Smoke-test at $M = 1$; compare to SMF canonical
Session F	T3 / T4 / T5 / T6 queue killed; no FINDINGS from 100 m	Re-launch via `longfiber_burst_launcher.sh`; checkpoint/resume should recover from last save
Session H	12-run audit matrix never executed	Serialize launches behind heavy-lock after VM reset
Cross-session	7 session branches unmerged to `main`	User integration pass per Rule P7
Infrastructure	Rule P5 heavy-lock not respected by sibling sessions	Add mandatory wrapper + watchdog; policy change

13 Recommended Next Actions

Immediate (user decision required).

1. Decide on burst VM reset vs. wait. If reset, accept loss of in-flight state across all sessions and re-launch in sequence.
2. Decide on integration pass: which of the 7 session branches to merge to `main` now, which to block on missing results.

After VM recovery (physics path).

3. Run Phase 14.02 A/B sweep. If pure $\mathrm{tr}(H)$ penalty fails at saddles, switch to $\sum |\lambda_i|$.
4. Complete Session E Sweep 2 (14 remaining configs).
5. Run Session D protocol on Session E Pareto candidates #1–3 — this is the gate for any lab push.
6. Session F: relaunch T3–T6 via checkpoint-resume; produce 100 m FINDINGS.
7. Session H 12-run matrix; pick a project-default cost.

Follow-up phases.

8. Newton-on-reduced-subspace on Session E candidates — direct Hessian factorization at $\mathrm{N}_\varphi = 57$, expected -90 dB.
9. Session A multi-variable demo: measure whether joint $\{\varphi, A, E\}$ wins ≥ 0.5 dB over phase-only.
10. Session C MMF at $M \geq 2$: first production run after $M = 1$ byte-identity check.

Policy.

11. Enforce Rule P5 heavy-lock via mandatory wrapper.
12. Add CPU / memory watchdog on burst VM.
13. Downgrade "light run" budget to ≤ 2 cores when a heavy tenant is active.

Report scope. This synthesis covers work committed to `main` and to the seven session branches `sessions/{A-multivar, C-multimode, D-simple, E-sweep, F-longfiber, H-cost}` as of 2026-04-17 compilation time. Branch `sessions/B-handoff` (documentation / handoff session) is not covered per user instruction.

Data traceability. Every numeric claim in this report is traceable to a file path in the codebase or on a session branch. Primary sources: `results/raman/phase13/FINDINGS.md`, `results/raman/phase15/benchmark.md`, `.planning/phases/14-.../14-01-SUMMAR` `.planning/phases/16-sweep-simple-profiles/16-SUMMARY.md`, `results/raman/phase_sweep_simple/candidates.md`, `sessions/D-simp` `sessions/F-longfiber:.planning/sessions/F-longfiber-status.md`.

Not in this report. (i) Long-form research briefs in `.planning/notes/*-research.md` — these remain the authoritative source on time-window derivations, MI thresholds, and optimizer choices. (ii) Raw JLD2 payloads (phases 13–17) — referenced but not embedded; load via `JLD2.@load`. (iii) Session B (handoff / documentation) — excluded per user instruction.